Proportional Logic Output:-

```
=== Truth Table checking KB |= R ===
   P     |    Q     |    R     |   Q -> P   |  P -> ~Q   |   Q v R    |    KB    |     R
-------------------------------------------------------------------------------------------------
 false   |  false   |  false   |    true    |    true    |   false    |  false   |    false
 false   |  false   |   true   |    true    |    true    |    true    |   true   |     true      <-- KB Model
 false   |   true   |  false   |   false    |    true    |    true    |  false   |    false
 false   |   true   |   true   |   false    |    true    |    true    |  false   |     true
  true   |  false   |  false   |    true    |    true    |   false    |  false   |    false
  true   |  false   |   true   |    true    |    true    |    true    |   true   |     true      <-- KB Model
  true   |   true   |  false   |    true    |   false    |    true    |  false   |    false
  true   |   true   |   true   |    true    |   false    |    true    |  false   |     true

KB entails R : True


=== Truth Table checking KB |= R -> P ===
   P     |    Q     |    R     |   Q -> P   |  P -> ~Q   |   Q v R    |    KB    |   R -> P
-------------------------------------------------------------------------------------------------
 false   |  false   |  false   |    true    |    true    |   false    |  false   |     true
 false   |  false   |   true   |    true    |    true    |    true    |   true   |    false      <-- KB Model
 false   |   true   |  false   |   false    |    true    |    true    |  false   |     true
 false   |   true   |   true   |   false    |    true    |    true    |  false   |    false
  true   |  false   |  false   |    true    |    true    |   false    |  false   |     true
  true   |  false   |   true   |    true    |    true    |    true    |   true   |     true      <-- KB Model
  true   |   true   |  false   |    true    |   false    |    true    |  false   |     true
  true   |   true   |   true   |    true    |   false    |    true    |  false   |     true

KB entails R -> P : False


=== Truth Table checking KB |= Q -> R ===
   P     |    Q     |    R     |   Q -> P   |  P -> ~Q   |   Q v R    |    KB    |   Q -> R
-------------------------------------------------------------------------------------------------
 false   |  false   |  false   |    true    |    true    |   false    |  false   |     true
 false   |  false   |   true   |    true    |    true    |    true    |   true   |     true      <-- KB Model
 false   |   true   |  false   |   false    |    true    |    true    |  false   |    false
 false   |   true   |   true   |   false    |    true    |    true    |  false   |     true
  true   |  false   |  false   |    true    |    true    |   false    |  false   |     true
  true   |  false   |   true   |    true    |    true    |    true    |   true   |     true      <-- KB Model
  true   |   true   |  false   |    true    |   false    |    true    |  false   |    false
  true   |   true   |   true   |    true    |   false    |    true    |  false   |     true

KB entails Q -> R : True
```